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Day - 4 (Networking Commands)

Date: 29/6/20

1. ping

2. Netstat

3. ifconfig

4. Traceroute vs tracepath

5. mtr

6. nslookup

7. ~~telnet~~ telnet

8. Hostname

9. ip

10. iwconfig

11. ss

12. arp

13. dig

14. nc (netcat)

15. whois

16. ifplugstatus

1. route

2. rmap

3. wget

4. watch

5. iptables

6. traceroute

7. curl vs wget

API endpoints
No call ~~using~~ k len

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Linux Networking Commands for DevOps beginners:

These commands are used by DevOps engineers to troubleshoot network issues, check servers, test ports, verify DNS, and inspect network interfaces.

Note: ~~You must~~ #

1. Connectivity checking commands:

These commands help you check whether a server, website, or IP address is reachable.

Ping:

Purpose: checks if another server or website is reachable.

Syntax: ping google.com

ping -c 4 ~~8.8.8.8~~ google.com

[-c 4 → send only 4 packets.]

[-i 2 → wait 2 seconds b/w packets]

[-w 3 → wait 3 seconds for reply]

Explanation:

ping sends small packets to a host and waits for a reply.

Example:

ping -c 4 8.8.8.8

This checks if google DNS is ~~not~~ reachable.

DevOps use:

used as the first command when checking internet or server connectivity.

tracert:

Purpose: Shows the path packets take to reach a destination.

Syntax:

tracert google.com

Explanation: It shows all network hops between your system and the destination.

DevOps use: used when a website or server is slow or unreachable, to find where the problem is happening.

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tracpath:

purpose: Similar to traceroute, but simpler

Syntax:

tracpath google.com

Explanation:

It shows the route to the destination and can also detect MTU issues.

Traceroute vs tracpath

Command

use

Beginner Note

→ traceroute

Detailed route checking

more options

→ tracpath

Simple route checking

Easier and usually no root needed.

mtr: → (my traceroute)

purpose: Combines ping and traceroute.

Syntax:

mtr google.com

-x → give me a one time report.

mtr -x -c 10 google.com

-c 10 → send exactly 10 probes.

Explanation:

It continuously checks each network hop and shows packet loss and latency.

Devops use:

Very useful for checking network delay, packet loss, or unstable connections.

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2. Network interface and IP commands:

These commands help you check IP addresses, network cards, and routes.

ip:

purpose: Modern commands to check and manage network interfaces, IP addresses, routes, and ARP entries.

Syntax:

ip addr
ip link
ip route
ip neigh

Examples:

ip addr

Shows IP addresses:

ip route

Shows routing table and default gateway:

ip link

Shows network interfaces.

DevOps use:

This is one of the most important networking commands in Linux.

ifconfig:

purpose: Shows network interface information.

Syntax:

ifconfig
ifconfig eth0 → eth0

Explanation:

It shows IP address, MAC address, and network interface status.

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modern replacement:
ip addr

Devops uses

ifconfig is old, in professional Devops work, use ip more often.

hostname:

purpose: shows or changes the system ~~host~~ hostname.

Syntax:

hostname

hostname -I

hostnamectl

Examples:

hostname

Shows system name.

hostname -I

Shows system IP address

sudo hostnamectl set-hostname web-server-01

changes the hostname.

Devops use:

useful for ~~identifying~~ identifying servers in logs, monitoring, cloud systems, and clusters.

iwconfig:

purpose: Shows wireless network interface information.

Syntax:

iwconfig

iwconfig wlan0

Explanation:

used mainly for Wi-Fi interfaces

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Devops use:
useful for laptops, Raspberry Pi, wireless servers, or edge devices.

3. ports and Services commands:

These commands help you check which services are running and which ports are open, and whether a server is accepting connections.

ss - socket statistics:

purpose:

ss is used to show socket information, listening ports, and active network connections.

Syntax:

ss -tulnp

Options:

- t → TCP connections
- U → UDP connections
- l → Listening ports
- n → show numeric ports number
- p → show process name and PID

Example:

ss -tulnp

Explanation:

This command shows all TCP and UDP services that are currently listening on the system.

Devops use:

Devops engineers use ss to check whether services like SSH, Nginx, Apache, MySQL, Docker or PostgreSQL are running and listening on the correct ports.

netstat → network statistics:

purpose: netstat shows network connections, listening ports, routing tables, and network statistics.

Syntax:

netstat -tulnp
netstat -rn

options:

- t → TCP connections
- U → UDP connections
- l → Listening ports
- n → Show numeric addresses and ports
- P → Show process name and PID
- r → Show routing table.

Example:

netstat -tulnp

Explanation:

~~net~~ This command shows services that are listening on TCP and UDP ports.

Devops Note:

netstat is an older command. In modern Linux system, ss is preferred.

nc → Netcat:

purpose: nc, also called Netcat, is used to test TCP or UDP connections and check whether a port is open.

Syntax: nc -zv host port

Example: nc -zv google.com 443

Explanation: This checks whether port 443 is open on google.com.

nc → netcat
-z → scan only, no data sent
-v → verbose output
443 → HTTP port.

HTTP → 80
HTTPS → 443

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Options:

- Z → Scan without sending data
- V → Verbose output.

Devops use:

Devops engineers use nc to test service connectivity, firewall rules, and open ports.

telnet: → Telecommunication Network:

Purpose: telnet is an old command used to connect to remote systems. Today, it is mostly used for simple port testing.

Syntax:

telnet host port

Example:

telnet google.com 80

Explanation:

This checks whether port 80 is reachable on google.com

Devops note:

Don't use telnet for secure remote login because it is not encrypted. Use ssh for login. For port testing, nc is usually better.

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4. DNS Troubleshooting Commands:

These commands help you check domain name resolution.

nslookup: → Name Server lookup

Purpose: checks DNS records of a domain

Syntax: nslookup google.com

Example: nslookup google.com

Shows the IP address of the domain

Devops use:

used to check whether DNS is resolving correctly.

dig: → Domain Information Groper:

Purpose: Advanced DNS lookup command.

Syntax:

dig google.com → checks default DNS record, usually A record

dig google.com A → finds IPv4 address

dig google.com mx → finds mail servers

dig google.com NS → finds name servers

dig -x 8.8.8.8 → performs reverse DNS lookup: IP to domain

Examples:

dig google.com -short

Shows only the IP address.

dig google.com mx

Shows mail servers records

Devops use:

used for serious DNS troubleshooting.

nslookup vs dig

Command

Use

Beginner Note

nslookup

Basic DNS checking

Simple

dig

Detailed DNS checking

more professional.

5. Local Network Commands:

These commands help you check local network information.

arp: → address Resolution protocol:

purpose:

Shows IP-to-MAC address mapping

Syntax:

`arp -a`

Explanation:

ARP maps an IP address to a MAC address in the local network.

modern replacement:

`ip neigh`

Devops use:

used when troubleshooting local network issues.

ifplugstatus:

purpose:

checks whether a network cable is plugged in.

Syntax:

`ifplugstatus`

`ifplugstatus eth0`

Example output:

`eth0: Link beat detected detected`

This means cable/link is connected.

`eth0: unplugged`

This means cable/link is disconnected.

Devops uses useful for checking physical ethernet connection.

6. Domain information Command:

Whois:

purpose: Shows domain registration information.

Syntax:

whois example.com

Explanation:

It can show register, nameservers, domain owner info, and expiry date.

DevOps use:

used to check domain ownership, expiry, and DNS registration details.

Important Modern Replacements:

old command	modern command	purpose
ifconfig	ip addr	check IP/interface
netstat	ss	check ports/sockets
arp	ip neigh	check ARP table
nslookup	dig	DNS troubleshooting
telnet	nc	port testing

Beginner DevOps Troubleshooting Flow:

When network is not working, follow this order:

hostname

ip addr

ip route

ping 8.8.8.8

ping google.com

dig google.com

ss -twnp

nc -zv server.com 443

traceroute google.com

mttr google.com

Meaning of the Flow:

Step	Command	Purpose
1.	hostname	check which server you are on
2.	ip addr	check ip address
3.	ip route	check gateway/default route
4.	ping 8.8.8.8	check internet by IP
5.	ping google.com	check DNS + internet
6.	dig google.com	check DNS details
7.	ss -tlnp	check listening services
8.	nc -zv host port	Test specific port
9.	tracert	check network path
10.	mtr	check packet loss and latency.

Simple Memory Trick:

Remember this order:

IP → Ping → DNS → Port → Path

Meaning:

ip addr = Do I have an IP?

ping = Can I reach the server?

dig / nslookup = Is DNS working?

ss / nc = Is the port open?

tracert / mtr = Where is the network problem?

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1. route:

route is used to view or manage the system routing table. it shows where network traffic will go.

Syntax:

route

route -n

Examples:

route -n

Shows routing table with IP addresses instead of names

sudo route add default gw 192.168.1.1

Add default gateway.

Modern command:

ip route

DevOps Use:

To check default gateway, routing issues, and network path problems.

2. nmap:

nmap is used for network scanning. It checks open ports, services, and live hosts.

Syntax:

nmap [options] target

Examples:

nmap 192.168.1.1

Scans common ports on one host

nmap -sV 192.168.1.1

Detects service versions

nmap -p 22, 80, 443. 192.168.1.1

Scans specific ports

nmap -sn 192.168.1.0/24

Find live devices in a network

DevOps Use:

To check if server ports like SSH, HTTP, HTTPS are open.

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3. Wget

wget is used to do download files from the internet.

Syntax:

wget URL

Examples:

wget https://example.com/file.zip

Downloads a file.

wget -o newfile.zip https://example.com/file.zip

Downloads and saves with a new name.

wget -c https://example.com/file.zip

Continues an incomplete download.

Devops use:

To download packages, scripts, backups, or installation files on servers.

4. Watch

Watch runs a command repeatedly and shows live output.

Syntax:

watch command

Examples:

watch date

Shows updated time every 2 seconds.

watch df -h

monitors disk space

watch free -m

monitors memory usage

watch -n 1 uptime

Runs commands every 1 second

Devops use:

To monitor server resources etc.

5. iptables:

iptables is used to manage Linux firewall rules.

1 Syntax:

```
sudo iptables [options]
```

Examples:

```
sudo iptables -L
```

Lists firewall rules.

```
sudo iptables -L -n -v
```

Shows detailed rules with IPs and packet count

```
sudo iptables -A INPUT -p tcp --dport 22 -j ACCEPT
```

Allows SSH port 22.

```
sudo iptables -A INPUT -p tcp --dport 80 -j ACCEPT
```

Allows HTTP port 80.

```
sudo iptables -A INPUT -p tcp --dport 443 -j ACCEPT
```

Allows HTTP port 443.

```
sudo iptables -A INPUT -p tcp --dport 8080 -j DROP
```

2 Block port 8080.

DevOps use:

To allow or block server traffic using firewall rules.

6. traceroute:

traceroute shows the path packets take from your system to a destination.

Syntax:

```
traceroute domain_or_ip
```

Examples:

```
traceroute google.com
```

Shows path to google

```
traceroute 8.8.8.8
```

Shows path to google DNS.

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Install if missing:

sudo apt install traceroute

Debugs use:

To troubleshoot network delay, packet path, and unreadable servers.

7. curl vs wget:

Both are used to transfer data from URLs, but they are used differently.

Command	Main use
curl	Testing APIs, HTTP requests, headers
wget	Downloading files

Curl:

curl https://example.com

Shows page/API response in terminal

curl -I https://example.com

Shows only headers

curl -X GET https://api.example.com

Sends GET request.

~~curl -X GET POST~~

API downloads
/jq for better output

curl -X POST -d "name=ali" https://example.com/api

Sends POST request

Wget:

wget https://example.com/file.zip

Downloads file

wget -c https://example.com/file.zip

Resume downloads.

for download
wget

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Easy Difference:

curl = view / test data

wget = download files

Devops use:

use curl for API testing and server response checking

Use wget for downloading files on linux server

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